MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY-EMERGENCY REPORT SYSTEM

1. SCOPE

1.1 This specification covers the detailed requirements for an amplifier used to amplify audio frequency signals to drive a group of loudspeakers in an emergency report system onboard a submarine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (D.C. Aluminum Electrolytic, Polarized) General Specification for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Tantalum Foil and Sintered-Slug) General Specification for.
MIL-A-20222 - Announcing-Equipment, Shipboard.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.
MIL-S-23012 - Submarine Intercommunications System AN/WIC-(V).

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The amplifier shall be in accordance with the applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.1.1 The equipment complement shall consist of two amplifiers housed in one enclosure.

3.2 Equipment function. The function of the equipment described herein shall be to provide the amplification of voice signals required for emergency report system circuit 4MC.

3.2.1 Circuit 4MC function. The function of the emergency report system, circuit 4MC, is to provide one way transmission of communications of an emergency nature from stations located throughout the ship to the ship's control center and to the maneuvering area.

3.2.2 Other components used in circuit 4MC but not covered by this specification are as specified in 3.2.2.1 and 3.2.2.2.

3.2.2.1 Emergency report switches. Emergency report switches, type AN/SIA-718A in accordance with MIL-S-23012 shall be used to transfer sound powered telephones from their normal circuit to circuit 4MC at input stations. Operation of any of these switches shall energize the amplifier.

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3.2.2.2 Loudspeakers. Types IC/SAA and IC/SBA loudspeakers in accordance with MIL-A-20222 shall be used at all locations using loudspeakers.

3.3 Enclosure. The enclosure shall be of drip-proof construction designed for bulkhead mounting. Amplifiers shall be drawout type mounted on guide rails and connected by male and female connectors. Provision shall be made for front servicing of all components. Dowels and matching bushings shall be provided for support and alignment of units.

3.4 Size and weight. Maximum overall dimensions of the enclosure shall be 12 inches in width, 15 inches in height, and 12 inches in depth. Minimum practical dimensions consistent with design requirements will be acceptable. The weight of the unit shall not exceed 40 pounds.

3.5 Semi-conductors. Semi-conductor circuits shall be used exclusively. Semi-conductors and rectifiers shall be approved types listed in MIL-STD-701.

3.6 Capacitors. Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in amplifier assemblies. Plug-in type electrolytics in accordance with MIL-C-62 may be utilized in the power supply.

3.7 Relays. Relays may be hermetically sealed.

3.8 Printed wiring. Printed wiring shall be in accordance with standoff type terminations as specified in MIL-STD-275.

3.9 Wiring. Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.10 Output power. The rated output power of each amplifier shall be 10.0 watts minimum.

3.11 Input facilities. Terminal blocks shall be provided to accommodate 4 input circuits. Individual emergency report station input circuits shall be paralleled external to the amplifier enclosure to limit the number of input circuit terminals required.

3.12 Power supply. Normal power input shall be from 115 volt, 60 cycle, single phase ship's service. A sintered plate nickel cadmium battery shall be provided as an emergency supply in the event of failure of the normal supply. A box for mounting the battery exterior to the amplifier enclosure shall be provided.

3.13 Active circuit. One amplifier shall be in the active circuit. The other amplifier shall be considered as a spare. The spare amplifier shall be used as a replacement module in the event of failure of the active module.

3.14 Test facilities. Test facilities shall be provided to test the amplifier in the spare position. A voltmeter shall be provided to measure the output of the amplifier. A test jack shall be provided to permit use of a microphone to insert a test signal to measure amplifier output.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 5830-N022)

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